TASK: 1

List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4 and IPv6 are versions of the Internet Protocol used to identify devices on a network. IPv6 was developed to provide a much larger address space than IPv4.

IPv4	IPv6
1. Uses a 32-bit address.	Uses a 128-bit address.
2. Supports approximately 4.3 billion addresses.	Supports an extremely large number of addresses (approximately 3.4×10^{38}).
3. Written in decimal format, such as `192.168.1.10`.	Written in hexadecimal format, such as `2001:db8::10`.
4. Uses four numbers separated by dots.	Uses eight groups of hexadecimal digits separated by colons, with abbreviation rules available.
5. Supports broadcast communication.	Does not use broadcast; it supports multicast and anycast.
6. NAT is commonly used to conserve public IPv4 addresses.	NAT is generally not required for address conservation because of the large address space.
7. ARP is used to discover link-layer addresses on IPv4 local networks.	Neighbor Discovery Protocol (NDP) is used for neighbor and link-layer address discovery.
8. IPsec support is available but is not mandatory in all implementations.	IPsec support is part of the IPv6 standards, although its use is not mandatory for every connection.

Conclusion: IPv4 is widely used today, while IPv6 provides a much larger address space and other improvements for modern networks.

TASK: 2

Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Given Network: `192.168.10.0/24`

Subnet Mask: `255.255.255.0`

Default Gateway: `192.168.10.1`

The following static IP addresses are assigned to four devices connected to the home Wi-Fi network.

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Smartphone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Explanation:

Each device has a unique IP address within the `192.168.10.0/24` network. All devices use the same subnet mask and default gateway. The default gateway allows them to communicate with devices outside their local network, such as internet servers.

Conclusion: Assigning unique static IP addresses helps identify devices and makes network management easier.

TASK: 3

Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Calculating Subnets and Hosts Using a /27 CIDR Mask

Given:

- Network: `192.168.1.0/24`
- Required subnet mask: `/27`
- Subnet mask: `255.255.255.224`

Step 1: Calculate the Number of Subnets

The original network is `/24`, and the new subnet prefix is `/27`.

Bits borrowed = 27 − 24 = **3 bits**

Number of subnets = 2^3 = **8 subnets**

Step 2: Calculate the Number of Usable Hosts

Host bits = 32 − 27 = **5 bits**

Total addresses per subnet = 2^5 = 32

Usable host addresses = 32 − 2 = **30 hosts per subnet**

Two addresses are reserved for the network address and broadcast address.

Step 3: List the First Three Subnets

The block size is 256 − 224 = **32**.

Subnet	Network Address	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.1.0/27	192.168.1.1	192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33	192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65	192.168.1.94	192.168.1.95

Step 4: List the Usable Host IP Ranges

Subnet 1:

- Network address: `192.168.1.0`
- Usable host IPs: `192.168.1.1 – 192.168.1.30`
- Broadcast address: `192.168.1.31`

Subnet 2:

- Network address: `192.168.1.32`
- Usable host IPs: `192.168.1.33 – 192.168.1.62`
- Broadcast address: `192.168.1.63`

Subnet 3:

- Network address: `192.168.1.64`
- Usable host IPs: `192.168.1.65 – 192.168.1.94`
- Broadcast address: `192.168.1.95`

Final Answer:

- Number of subnets: **8**
- Usable hosts per subnet: **30**
- Total usable host addresses across all 8 subnets: **240**

TASK: 4

**You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.**

Finding the Network Address, Broadcast Address, and Valid Host Range

Given:

- IP Address: `10.0.5.17`
- Subnet Mask: `255.255.255.0`

Step 1: Convert the Subnet Mask to CIDR

The subnet mask `255.255.255.0` contains 24 network bits.

Therefore, the CIDR notation is **/24**.

Step 2: Calculate the Network Address

IP address: `10.0.5.17`

Subnet mask: `255.255.255.0`

The first three octets represent the network portion, while the last octet represents the host portion.

Network Address: `10.0.5.0`

Step 3: Calculate the Broadcast Address

For a `/24` subnet, the last octet ranges from 0 to 255.

Broadcast Address: `10.0.5.255`

Step 4: Calculate the Valid Host Range

The network address and broadcast address cannot be assigned to ordinary hosts.

Therefore:

Item	Result
Given IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR Notation	/24
Network Address	10.0.5.0
First Usable Host	10.0.5.1
Last Usable Host	10.0.5.254
Broadcast Address	10.0.5.255
Total Usable Hosts	254

Conclusion: The IP address `10.0.5.17` belongs to the `10.0.5.0/24` network, which supports 254 usable host addresses.

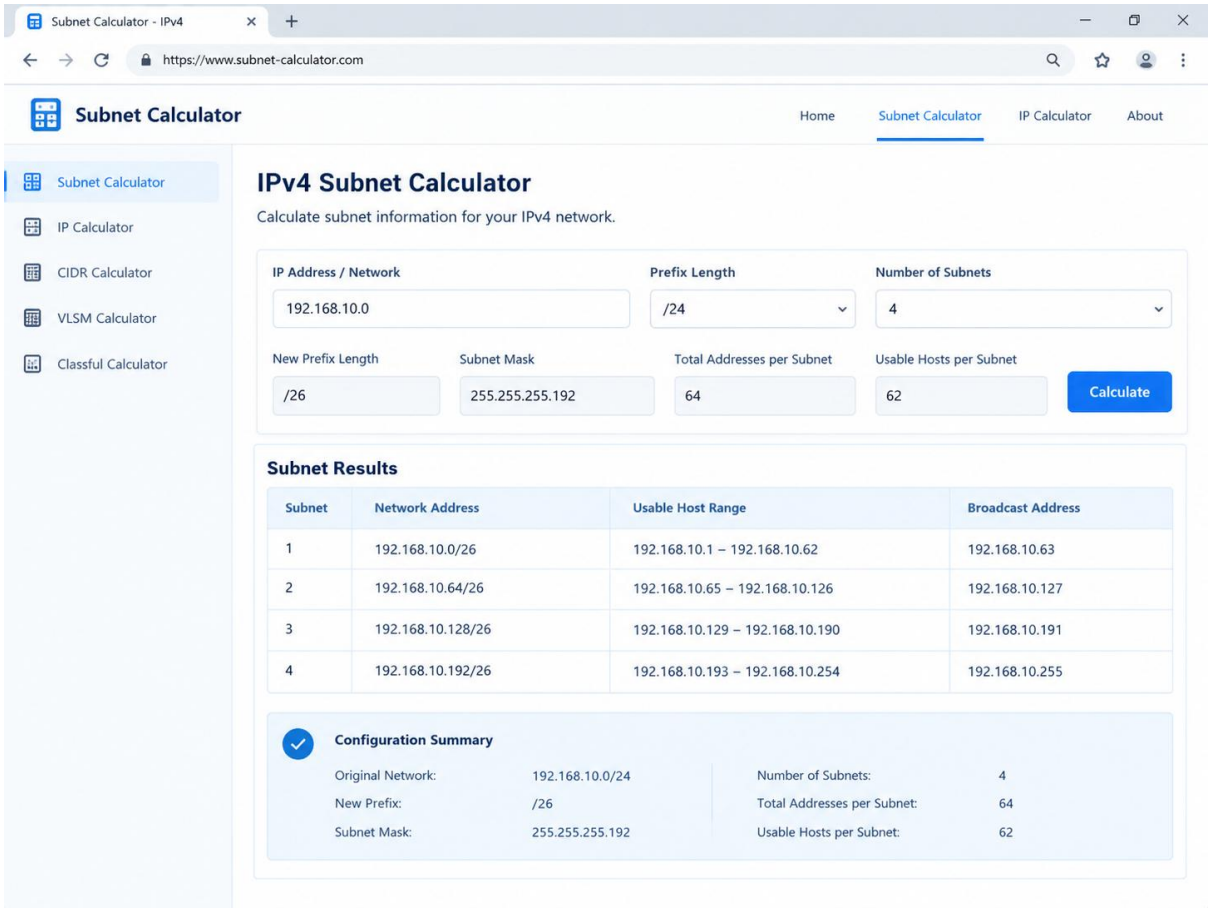
TASK: 5

Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Tool Used: Subnet Calculator

Steps Performed:

1. Opened the online subnet calculator.
2. Entered the IP address `10.0.5.17`.
3. Selected the subnet mask `255.255.255.0 (/24)`.
4. Checked the network address, broadcast address, and valid host range.
5. Compared the results with my manual calculations.



Results:

Parameter	Calculated Result
IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Usable Host	10.0.5.1
Last Usable Host	10.0.5.254
Total Usable Hosts	254

Verification Statement: My manual subnet calculation matched the results shown in the online subnet calculator.